Dheeraj Kumar

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Education

Bihar Engineering University (BEU) Aug 2024 - June 2028
Bakhtiyarpur College of Engineering, Patna ***2nd Year***
B.Tech in Computer Science & Engineering (IoT Specialization)

R.D.S College, Muzaffarpur 2021 - 2023
Intermediate - Science Stream (PCM)

Primus Public School, Muzaffarpur
Matriculation (Class X)

Skills

Programming Languages: Python, SQL,C, C++

Data Science & ML: NumPy, Pandas, Matplotlib, Seaborn, Scikit-learn, Time-Series Analysis

Web & Frameworks: Flask, FastAPI, Streamlit, React, REST APIs

Tools & Technologies: Git, GitHub, Docker, MySQL, Tableau, Jupyter Notebook, VS Code, Render

Project Work

Green Fleet Optimizer — Quantum-Inspired Fuel Optimization Live Demo | GitHub

- Developed an AI-driven fleet optimization system combining ML model and Quantum Particle Swarm Optimization (QPSO) to minimize emissions and deployment costs
- Built a robust REST API using Flask to process complex optimization algorithms and serve physics-based fuel predictions
- Created an interactive frontend dashboard using React and Recharts for real-time visualization of optimization convergence and CO_2 footprints
- **Tech Stack:** Python, Flask, React, Advanced grading boosting model, QPSO, Recharts, REST API

Laptop Price Estimator Live Demo | GitHub

- Built an end-to-end machine learning pipeline using Scikit-learn to predict laptop prices based on hardware specifications
- Performed comprehensive EDA and feature engineering using Pandas and NumPy, including log-transformations and extracting PPI from screen resolution
- Developed and deployed an interactive web application using Streamlit to provide real-time price estimations based on user inputs
- **Tech Stack:** Python, Scikit-learn, Pandas, NumPy, Matplotlib, Seaborn, Streamlit, ML Pipelines, Render (Cloud Deployment), Git/GitHub

Quantitative Research Job Simulation (JPMorgan Chase & Co.) GitHub

- Developed a natural gas pricing model using Linear Regression to extrapolate future prices and account for seasonal monthly fluctuations
- Built a retail credit risk model using Logistic Regression to predict Probability of Default (PD) and calculate financial Expected Loss
- Implemented a Dynamic Programming algorithm for optimal FICO score quantization to maximize Log-Likelihood in automated quoting systems
- **Tech Stack:** Python, Scikit-learn, Pandas, NumPy, Matplotlib, Seaborn, ydata-profiling, Time-Series Modeling, Dynamic Programming, Financial Modeling

Certifications and Achievements

- **JPMorgan Chase & Co. - Quantitative Research Job Simulation:** Completed virtual simulation on quantitative finance, risk analysis, and pricing models
- **Problem Solving:** Solved 300+ coding problems on LeetCode in C++ focusing on data structures and algorithms

Areas of Interest

- Artificial Intelligence | Machine Learning | Software Engineering | Data Science